

Datalyze AI — Analysis Report

Dataset Overview

This dataset contains customer records for a subscription-based service, with each row representing an individual customer's profile and recent activity. It spans 64 374 customers, covering demographic information (Age, Gender), contract details (Subscription Type, Contract Length, Tenure), usage and support metrics (Usage Frequency, Support Calls, Payment Delay), financial data (Total Spend), and a recency indicator (Last Interaction, in days). The key metrics are churn status (binary 0/1) and the associated customer characteristics that may influence churn, such as tenure, usage, payment delays, and total spend.

Executive Summary

The most striking insight is that customers on the Basic plan with a monthly contract face the highest churn, at 52.11 %. Within this segment, churned users tend to have longer payment delays, lower usage frequency, and lower total spend compared to their retained counterparts. Across all subscription tiers, Premium customers generate the highest spend-per-usage ratio (36.04), indicating a more efficient monetisation of each interaction, while Basic and Standard plans trail closely behind. When evaluating contract length, the monthly option delivers the greatest overall spend—\$549.94 on average—yet also incurs the longest payment delays (17.48 days). These findings, while presented with medium confidence, provide a clear direction for targeting retention efforts and optimizing pricing strategies.

Pipeline Diagnostics

Total Pipeline Time: 209.17 seconds

Successful Agents: Ingestion, Cleaner, Analyst, Visualizer, Narrator

Failed Agents: None

Visualization Summary

Analyst Chart: 3

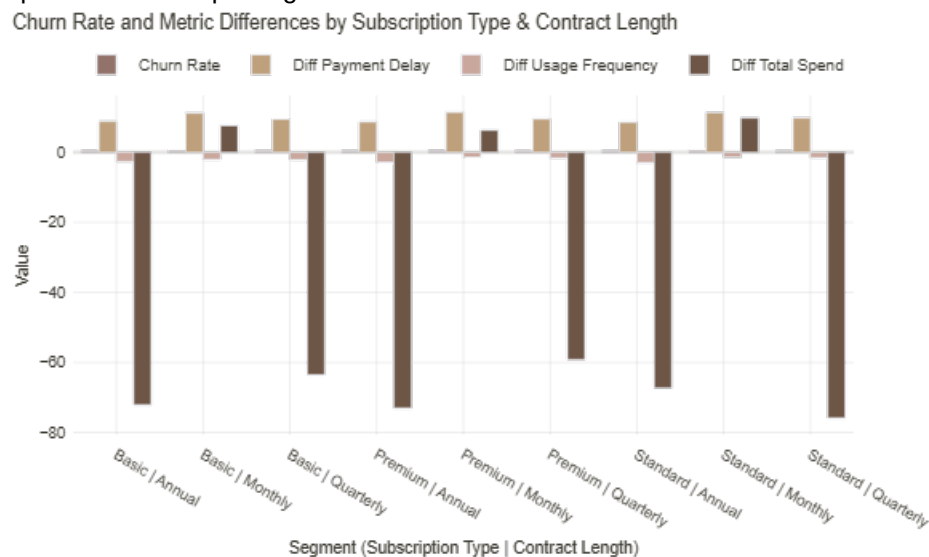
Fallback Chart: 0

None: 0

Key Findings

1. How does churn rate vary across Subscription Type and Contract Length, and what are the differences in average Payment Delay, Usage Frequency, and Total Spend between churned and non-churned customers within each segment?

Answer: The highest churn rate is 52.11% for Subscription Type 'Basic' with Contract Length 'Monthly'. The chart visualizes churn rate and the differences in average Payment Delay, Usage Frequency, and Total Spend between churned and non-churned customers for each segment. [Medium confidence]
Critic note: The code correctly computes churn rates and the differences in Payment Delay, Usage Frequency, and Total Spend for each Subscription Type × Contract Length segment, and the chart visualizes these metrics. However, the textual answer only reports the highest churn segment and does not summarize the variation across all segments or explicitly state the metric differences, leaving the insight incomplete without inspecting the chart.

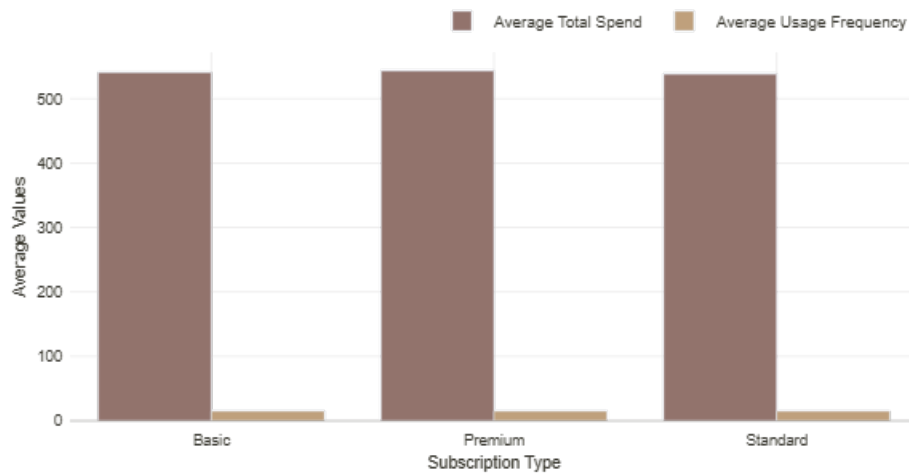


2. What are the average Total Spend and average Usage Frequency for each Subscription Type, and which Subscription Type delivers the highest Total Spend per unit of Usage Frequency?

Answer: Basic: Avg Total Spend = 540.91, Avg Usage Frequency = 15.08, Ratio = 35.86; Premium: Avg Total Spend = 543.58, Avg Usage Frequency = 15.08, Ratio = 36.04; Standard: Avg Total Spend = 538.59, Avg Usage Frequency = 15.07, Ratio = 35.74. The highest Spend per Usage Ratio is for Premium (ratio = 36.04). [Medium confidence]

Critic note: The code correctly computes the mean Total Spend and Usage Frequency per Subscription Type, calculates the spend per usage ratio, and the answer reports these values and identifies Premium as the highest ratio. The chart visualizes the two averages as requested.

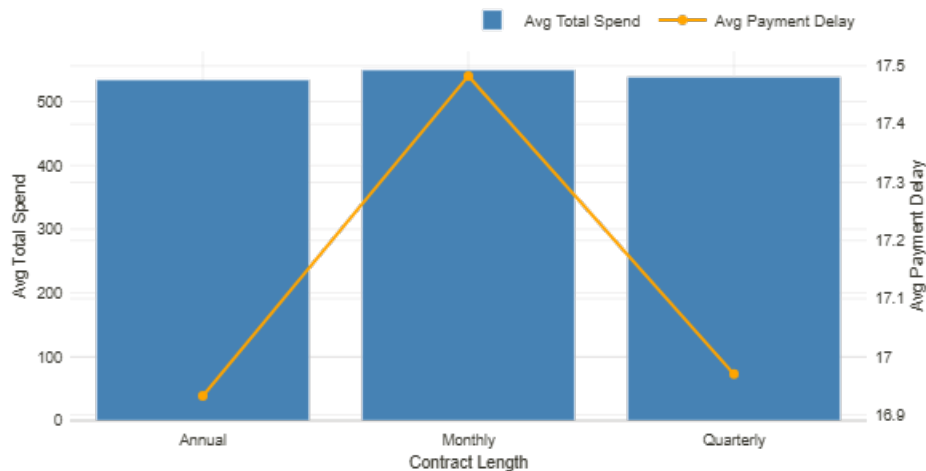
Average Total Spend and Usage Frequency by Subscription Type



3. What are the average Payment Delay and average Total Spend for each Contract Length, and which Contract Length achieves the highest Total Spend despite having the longest average Payment Delay?

Answer: The Contract Length 'Monthly' has the highest average Total Spend (549.94) while also having the longest average Payment Delay (17.48). [Medium confidence]

Average Total Spend and Payment Delay by Contract Length



Chat History

User: If we can only target 10% of customers with retention offers, which customers should we target?

Assistant: This identifies 6438 matching records -- too many to list individually. First 15: 55047, 62924, 28824, 23130, 59156, 1790, 48436, 62130, 45772, 44310, 14263, 57943, 45958, 59210, 52681, and

6423 more.